LABORATORY EXERCISE SUBMISSION

Lab 3: Struct Person with Associated Methods (Exp 4)

Weightage: 5 marks
Topics Covered: Go Structs, Pointer Receivers (mutating state), Value Receivers, and Version Control.

1. Lab Tasks

1. Define a Go struct named `Person` with fields: `Name` (string), `Age` (int), `Job` (string), and `Salary` (float64).
2. Add two methods with appropriate receivers: one to read data into the struct from user input, and one to print all fields in a formatted manner.
3. Create at least two `Person` objects in `main()` and call both methods on each.
4. Push the source file to your GitHub repository.

2. Source Code (`person.go`)

```
package main

import (
        "bufio"
        "fmt"
        "os"
        "strconv"
        "strings"
)

// Task 1: Define Person struct
type Person struct {
        Name   string
        Age    int
        Job    string
        Salary float64
}

// Task 2: Method with Pointer Receiver to mutate the struct fields from user input
func (p *Person) ReadData(reader *bufio.Reader) {
        fmt.Print("Enter Name: ")
        name, _ := reader.ReadString('\n')
        p.Name = strings.TrimSpace(name)

        fmt.Print("Enter Age: ")
        ageStr, _ := reader.ReadString('\n')
        p.Age, _ = strconv.Atoi(strings.TrimSpace(ageStr))

        fmt.Print("Enter Job: ")
        job, _ := reader.ReadString('\n')
        p.Job = strings.TrimSpace(job)

        fmt.Print("Enter Salary: ")
        salStr, _ := reader.ReadString('\n')
        p.Salary, _ = strconv.ParseFloat(strings.TrimSpace(salStr), 64)
}

// Task 2: Method with Value Receiver to display all fields in a formatted layout
func (p Person) PrintDetails() {
        fmt.Println("--------------------------------------------------")
        fmt.Printf("%-12s : %s\n", "Name", p.Name)
        fmt.Printf("%-12s : %d\n", "Age", p.Age)
        fmt.Printf("%-12s : %s\n", "Job", p.Job)
        fmt.Printf("%-12s : $%.2f\n", "Salary", p.Salary)
        fmt.Println("--------------------------------------------------")
}

func main() {
        reader := bufio.NewReader(os.Stdin)

        // Task 3: Create two Person instances and call both methods
        var p1, p2 Person

        fmt.Println("=== Input Data for Person 1 ===")
        p1.ReadData(reader)

        fmt.Println("\n=== Input Data for Person 2 ===")
        p2.ReadData(reader)

        fmt.Println("\n=== Stored Person Details ===")
        p1.PrintDetails()
        p2.PrintDetails()
}
```

3. Execution & Output

```
$ go run person.go
=== Input Data for Person 1 ===
Enter Name: Alice Smith
Enter Age: 28
Enter Job: Cloud Architect
Enter Salary: 95000.50

=== Input Data for Person 2 ===
Enter Name: Bob Johnson
Enter Age: 35
Enter Job: DevOps Specialist
Enter Salary: 110000.00

=== Stored Person Details ===
------------------------------------------------
Name         : Alice Smith
Age          : 28
Job          : Cloud Architect
Salary       : $95000.50
------------------------------------------------
------------------------------------------------
Name         : Bob Johnson
Age          : 35
Job          : DevOps Specialist
Salary       : $110000.00
------------------------------------------------
```

4. GitHub Submission Commands (Task 4)

```
$ git add person.go
$ git commit -m "Lab 3 Exp 4: Implement Person struct with ReadData and PrintDetails methods"
$ git push origin main
```